Experiment : 4

❖ **Aim :**
To design and develop a **Dynamic To-Do List Web Application** using **HTML, CSS, and JavaScript** that allows users to add, edit, delete, and manage tasks interactively.

❖ **Tasks :**

• **Project Setup & Introduction** – Create a project folder named **dynamic_todo_list** and add linked HTML, CSS, and JavaScript files with a short description of the application.

• **To-Do List Interface Design** – Create an input field to enter tasks, an **Add Task** button, and display tasks in a list format using HTML.

• **Styling with CSS** – Apply CSS styling to input fields, buttons, and task list to ensure proper alignment, spacing, and visual clarity.

• **Add & Display Tasks Using JavaScript** – Capture user input, add tasks dynamically to the list, and prevent empty tasks from being added.

• **Edit & Delete Task Features** – Provide **Edit** and **Delete** buttons for each task to allow modification and removal of tasks dynamically.

• **Task Completion Status (Bonus)** – Add a checkbox or toggle to mark tasks as completed and visually indicate completed tasks using a strikethrough effect.

❖ **Tech Stack Used :**
• **HTML** • **CSS** • **JavaScript**

❖ Html Code :

```
<!--
Project Title: Dynamic To-Do List
Name: Vansh
Experiment: 4
Date: 2026
-->

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Dynamic To-Do List</title>
<link rel="stylesheet" href="style.css">
```

```html
</head>
<body>

<header>
<h1>My To-Do List</h1>
<p>A simple to-do list to add, edit, delete, and complete tasks</p>
</header>

<main class="container">

<div class="input-area">
<input type="text" id="taskInput" placeholder="Enter a new task">
<button onclick="addTask()">Add Task</button>
</div>

<ul id="taskList">
</ul>

</main>

<footer>
<p>© 2026 Vansh</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

❖ **CSS Code :**

```
* {
box-sizing: border-box;
font-family: Arial, sans-serif;
}

body {
margin: 0;
background-color: white;
}

header {
background-color: rgb(15, 79, 199);
color: white;
padding: 20px;
text-align: center;
}

.container {
background-color: white;
max-width: 500px;
margin: 30px auto;
padding: 20px;
border-radius: 10px;
}

.input-area {
display: flex;
gap: 10px;
}

input {
flex: 1;
padding: 10px;
font-size: 14px;
```

```
}

button {
padding: 10px 15px;
background-color: blue;
color: white;
border: none;
cursor: pointer;
border-radius: 5px;
}

button:hover {
background-color: #3b55a0;
}

ul {
list-style: none;
padding: 0;
margin-top: 20px;
}

li {
background-color: whitesmoke;
padding: 10px;
margin-bottom: 10px;
border-radius: 6px;
display: flex;
align-items: center;
justify-content: space-between;
}

.completed {
text-decoration: line-through;
color: gray;
}
```

```
.task-buttons button {
margin-left: 5px;
font-size: 12px;
}

footer {
text-align: center;
padding: 10px;
color: black;
}
```

❖ **Javascript Code :**

```
function addTask() {
const input = document.getElementById("taskInput");
const taskText = input.value.trim();

if (taskText === "") {
alert("Please enter a task");
return;
}

const li = document.createElement("li");

const checkbox = document.createElement("input");
checkbox.type = "checkbox";
checkbox.onclick = () => toggleTask(li);

const span = document.createElement("span");
span.innerText = taskText;

const btnDiv = document.createElement("div");
btnDiv.className = "task-buttons";

const editBtn = document.createElement("button");
editBtn.innerText = "Edit";
editBtn.onclick = () => editTask(span);

const deleteBtn = document.createElement("button");
```

```
deleteBtn.innerText = "Delete";
deleteBtn.onclick = () => deleteTask(li);

btnDiv.appendChild(editBtn);
btnDiv.appendChild(deleteBtn);

li.appendChild(checkbox);
li.appendChild(span);
li.appendChild(btnDiv);

document.getElementById("taskList").appendChild(li);

input.value = "";
}

function deleteTask(task) {
task.remove();
}

function editTask(span) {
const newText = prompt("Edit your task:", span.innerText);
if (newText !== null && newText.trim() !== "") {
span.innerText = newText;
}
}

function toggleTask(task) {
task.classList.toggle("completed");
}
```

❖ **Output :**

My To-Do List

A simple to-do list to add, edit, delete, and complete tasks

Enter a new task | Add Task

- [ ] Do Web Dev Assignment-3 Edit Delete
- [ ] Make DBMS Lab file Edit Delete

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